PROJECT_FINAL_1

1.Introduction

1-0. Names: -

1. Kubra Banu
2. Joash Kawal

1-1. Summary of your question and findings

Kubra: The production efficiency of the companies in the industry electronics components not classified elsewhere have been impacted positively in turn leading to a positive stock market reaction due to the Event Chips Act. The Chips Act has enabled the production of semiconductors domestically, The electronics components industry provides components like resistors, capacitors, and others in the manufacturing of semiconductors which has led to their increased in production and value.

Joash: The Chips and Science Act promises good prospects for the United States Semiconductor and Related Devices sector. This is because the Chips Act aims to increase production within the US and will, therefore, have a positive influence on all companies in the semiconductor business. As a result, the Chips Act will improve the financial return on revenues.

2.Background

2-1. Overview of the event:

The chips and Science act which was enacted on 9th of august 2022 was done to create helpful incentives to produce semiconductors. The act authorizes roughly $280 billion in new funding to help in domestic research and manufacturing of the semiconductor in the United States. Its aim is to strengthen the supply chain resilience and as of March 2024 analyst estimated

that the act incentivized between twenty to fifty separate potential projects.
https://en.wikipedia.org/wiki/CHIPS_and_Science_Act

2-2. Overview of the industry:

Joash: Interested Industry: -

SIC Code - 3674

Industry - Semiconductors and Related Devices

Description of the interested industry: -

The Semiconductor Industry along with related devices including transistors, diodes, microprocessors, integrated circuits, and the like serve as the basis of any electronic device or component that makes up a computer device such as the CPU (Central Processing Unit) and the GPU (Graphics Processing Unit). The other sectors that benefit and utilize semiconductors and related devices include but are not limited to robotics, healthcare, automotive, and communication. It is a billion-dollar industry that consists of major players such as Intel, NVIDIA, and TSMSC, manufacturing semiconductors and chips that perform tasks that involve vast amounts of data being processed which are extremely relevant to the present and future years to come.

Resources - https://siccode.com/siccode/3674/semiconductors-devices

Selected Company with Edgar webpage: -

Intel Corp

Edgar webpage -
https://www.sec.gov/edgar/browse/?CIK=50863&owner=exclude

Description of selected company: -

Intel Corporation is a software company based in the US that manufactures semiconductors, chipsets, and CPUs (Central Processing Units) such as the Xeon processors that power the most advanced supercomputers. They are famously known for their rollout of the Core line of CPU which was and currently is popular among gaming enthusiasts to run high-end games which

require significant computing capability. The components are designed for commercial as well as business use cases wherein they are utilized for undertaking tasks in demanding applications, used in various fields such as entertainment, architecture, manufacturing design, engineering, and others. Intel Corporation initially started as a company that designed logic circuits, microprocessors, and memory chips and later went on to be one of the largest semiconductor manufacturers with an annual revenue of 55 billion as per the latest reports.

Resources - https://en.wikipedia.org/wiki/Intel

Kubra:

Interested industry's 4 digits SIC code and name: - **3679** is the SIC code and the interested industry is electronics components, not classified elsewhere.

Description of the interested industry: -

The industry named electronic components, not elsewhere classified under SIC (Standard industrial classification) code **3679** consists of a variety of electronic components that are generally used for various applications across different sectors such as aerospace, healthcare, telecommunication, and so on. Industries in general manufacture components such as connectors, antennas, resistors, Quartz, and various other components.

https://siccode.com/sic-code/3679/electronic-components

One of the companies in the interested industry and its SEC EDGAR page: -

Advanced Energy Industries, Inc

https://www.sec.gov/edgar/browse/?CIK=927003&owner=exclude

https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK=0000927003&owner=include&count=40&hidefilings=0

Description of the company selected (3-4 sentences)

Advanced Energy Industries, Inc is an American multinational technology company that has its headquarters in Denver, Colorado. It was founded in the year 1981 in Fort Collins, Colorado, USA by Doug Schatz. They have their manufacturing facilities not just in the USA but also in EMEA (Europe, the

middle east and Africa) and Asia. Advanced energy products include thermal measuring systems, remote plasma sources, power supplies and its largest customers are Advanced semiconductor materials, Tokyo Electron and more.

https://en.wikipedia.org/wiki/Advanced_Energy

2-3. Research questions with supporting evidence as well as logic.

Joash: The evidence or the link between the selected industry's performance & the event: -

The semiconductor industry's performance following The Chips and Science Act being enacted will significantly change, impacting how well the companies will fare in the coming years. Due to the nature of the act which enables manufacturing and fabrication to be carried out in the home country, companies that have already been manufacturing semiconductors domestically without relying on outsourcing their manufacturing to other countries are sure to benefit. Regarding outsourcing, NVIDIA, Intel Corp's competitor in the US, depends on TSMC, a Taiwan-based semiconductor manufacturer to fabricate the chips which ultimately run on the GPU's (Graphic Processing Unit) that NVIDIA sells commercially. The Chips and Science Act encourages funding companies that manufacture domestically as in the case of Intel. Recently, Intel received 8.5 billion in funding from the current government administration in hopes of being at the forefront of innovation in the world market regarding semiconductors and artificial intelligence which are directly related.

Resources - https://www.intel.com/content/www/us/en/newsroom/news/us-chips-act-intel-direct-funding.html

Hypothesis, Question & Economic Indicator:

The Chips and Science Act will positively impact the Semiconductor and Related Devices industry in the US.

Kubra:

The main aim of the Chips Act is to provide funding for the companies that manufacture semiconductors, so that the manufacturing can be done domestically, and they wouldn't have to rely on external sources. Even though the industry's electronics components, not classified elsewhere, don't directly come under the semiconductor manufacturing industry, they will benefit from

the Chips Act. Many companies in the industry of electronics components not elsewhere classify supply materials to the manufacturing of semiconductors, including resistors, capacitors and various other parts that are used in making the semiconductors as semiconductor manufacturing increases the need for the electronic components used in the manufacturing of the semiconductors will also increase. Due to the chips act the manufacturing of semiconductors is going to increase and the demand for the components or materials that are deemed necessary in the manufacturing of semiconductors will also increase and hence the industry electronics components not elsewhere classified is going to benefit from the chips act.

The following links are evidence of how the electronics components industry is related to the semiconductor industry:

https://en.m.wikipedia.org/wiki/Electronic_component

https://www.investopedia.com/terms/s/semiconductor.asp

These are the links of how they are going to benefit from the chips act :

https://crsreports.congress.gov/product/pdf/R/R47523

https://www.pwc.com/us/en/library/chips-act.html

https://www.mckinsey.com/industries/public-sector/our-insights/the-chips-andscience-act-heres-whats-in-it

https://www.nist.gov/system/files/documents/2024/04/19/Amended%20CHIPSCommercial%20Fabrication%20Facilities%20NOFO%20Amendment.pdf

Hypothesis: -

The production efficiency of the companies in the electronics components industry not elsewhere classified has been positively affected by the chips act, in turn leading to a positive stock market reaction.

3.Data

3-1 Brief description of compustat, CRSP and WRDS

Compustat and CRSP these two are the guide features of the database available in the WRDS. Compustat covers the accounts data, and it is separated into databases covering one section as a global database and the other as North America whereas the CRSP covers the price data for the companies listed in the United States.

Reference: https://medium.com/specialist-library-support/introduction-to-crsp-and-computstat-via-wrds-7c9b5b666c03

WRDS (Wharton Research Data Services) is a research platform to global institutions that provides the leading business intelligence and data analytics that enable you to have an insight into the latest innovation in research. They generally achieve this through partnering with different global vendors.

Reference: https://wrds-www.wharton.upenn.edu/pages/about/what-wrds/

3-2. Data preparation:

The format of required table for question 3-2

Step	Name of Data Frame	Description	# of obs.	List of fields
1	info_crsp	Downloading the data from CRSP. We use the data frame name info_crsp to form connection to the database and retrieve the desired data.	80789	'siccd', 'permno', 'ticker','comnam', 'cusip','namedt' these are the list of fields

2	info_crsp	Removing duplicates by key Permno : initially the total number of observations are 80786 and in them the number of duplicates are 43014 and once the duplicates have been dropped by keeping the last the observation is 37776	37776	Permno
3.	info_crsp	Showing the unique number of companies : once the duplicates have been dropped everything that remains unique and when we cross verify it confirms there are no duplicates.	37776	Permno
4.	Crsp_out	Sub setting permno and adding the event date.	37776	Permno, eventdate
5.	abcar	Downloading the abnormal return from WRDS	9281	Model, permno, evtdate, cret, car, bhar, nrets, nrets_est

6.	crsp	Merge both the data frame ,abcar and info_crsp, we use inner join to merge the data and the key is permno.	9281	Model, permno, evtdate, cret, car, bhar, nrets, nrets_est, siccd, ticker, comnam, cusip , namedt.
7	Crsp_comp	Merging the compustat data and crsp data frame, using inner join and the keys are permno and year	5667	Model , permno, evtdat, cret, car, bhar, nrets, nrets_est, siccd, ticker, comnam, cusip, namedt, year, sich, gvkey, datadate, conm, revt, ni,

3-3. Variable Definition

Information based on based on final data set crsp_comp

Variable name	Definition	# of missing values	Types
permno	PERMNO are unique permanent identifications numbers assigned by CRSP to all the companies that are listed in their dataset.	0	Int64
evtdate		0	Object
car	A cumulative abnormal return (CAR) is termed as the sum of all abnormal returns, and they are typically used to measure the effects of the events.	5	Float64
siccd	The Standard Industrial Classification (SIC) was a system for classifying industries by a four-digit code as a method of standardizing industry classification for statistical purposes across agencies	0	Int64
revt	The REV is not available for utility companies. Under the Financial Services Definition, this can be represented other revenue/income not classified elsewhere and not allocated to insurance, broker/dealer, or real estate operations. This item is a component of Income - Noninterest - Total (Bank) (INITB) and Revenue - Total (REVT).	400	Float64

log_revt	It is a statistical technique where the natural logarithm (log) of a company's revenue is calculated to make the data easier to analyze	420	Float 64
ni	This item represents the income or loss reported by a company after expenses and losses have been subtracted from all revenues and gains for the fiscal period including extraordinary items and discontinued operations. This item, for banks, includes securities gains and losses.	400	Float 64
semiconductor	A semiconductor is a material that is between the conductor and insulator in ability to conduct electrical current	0	Bool
Electronics	An electronic component is any basic discrete electronic device or physical entity part of an electronic system used to affect electrons or their associated fields.	0	Bool

References: -

1. Permno

https://libguides.stanford.edu/library/wrds/identifiers-linking/files

2. CAR

https://www.investopedia.com/terms/a/abnormalreturn.asp#:~:text=Abnormal%20returns%20can%20be%20produced,events%20have%20on%20stock%20prices

3. Semiconductor

https://en.wikipedia.org/wiki/Semiconductor

4. Electronics components are not classified elsewhere:

https://en.wikipedia.org/wiki/Electronic_component

5. Siccd:

https://en.wikipedia.org/wiki/Standard_Industrial_Classification

6. Revt, log_revt, ni

https://wrds-www.wharton.upenn.edu/pages/get-data/compustat-capital-iq-standard-poors/compustat/north-america-daily/fundamentals-annual/

4: - DATA ANALYSIS

4-1. Descriptive statistics

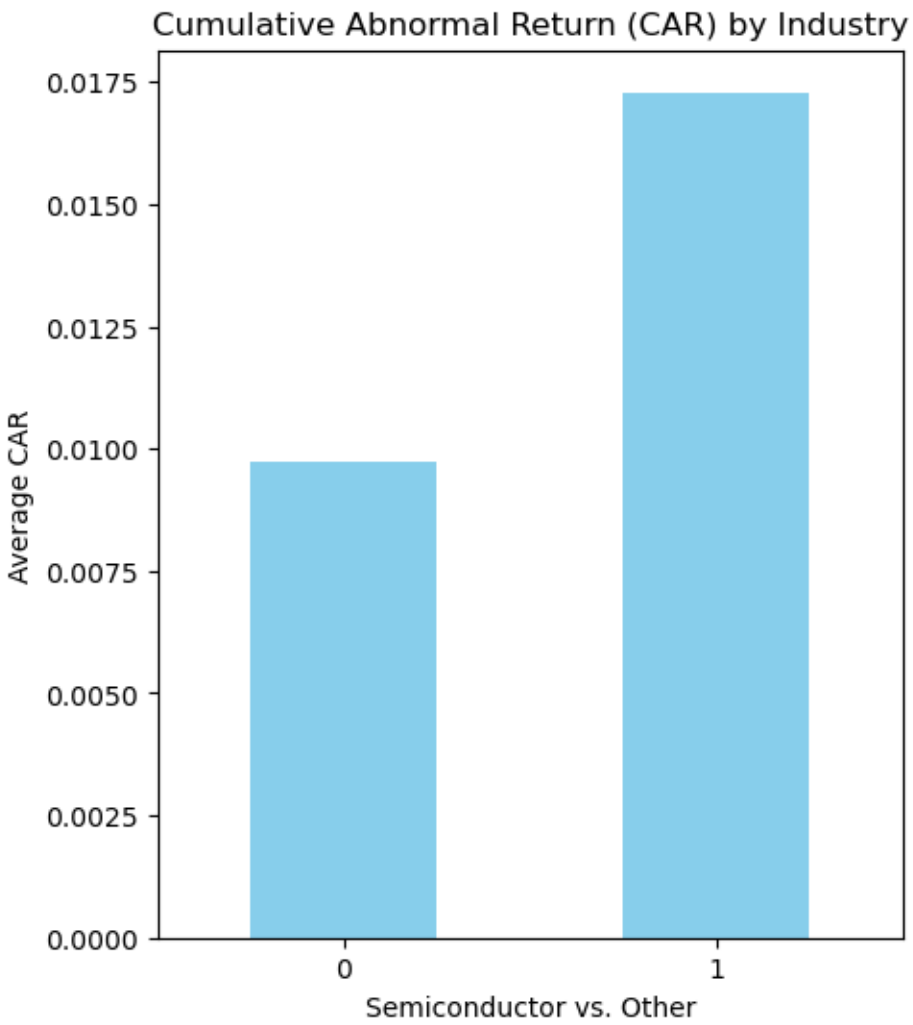
```
crsp_comp[['car', 'log_revt', 'semiconductor', 'electronics', 'revt']].describe() #descriptive statistics
```

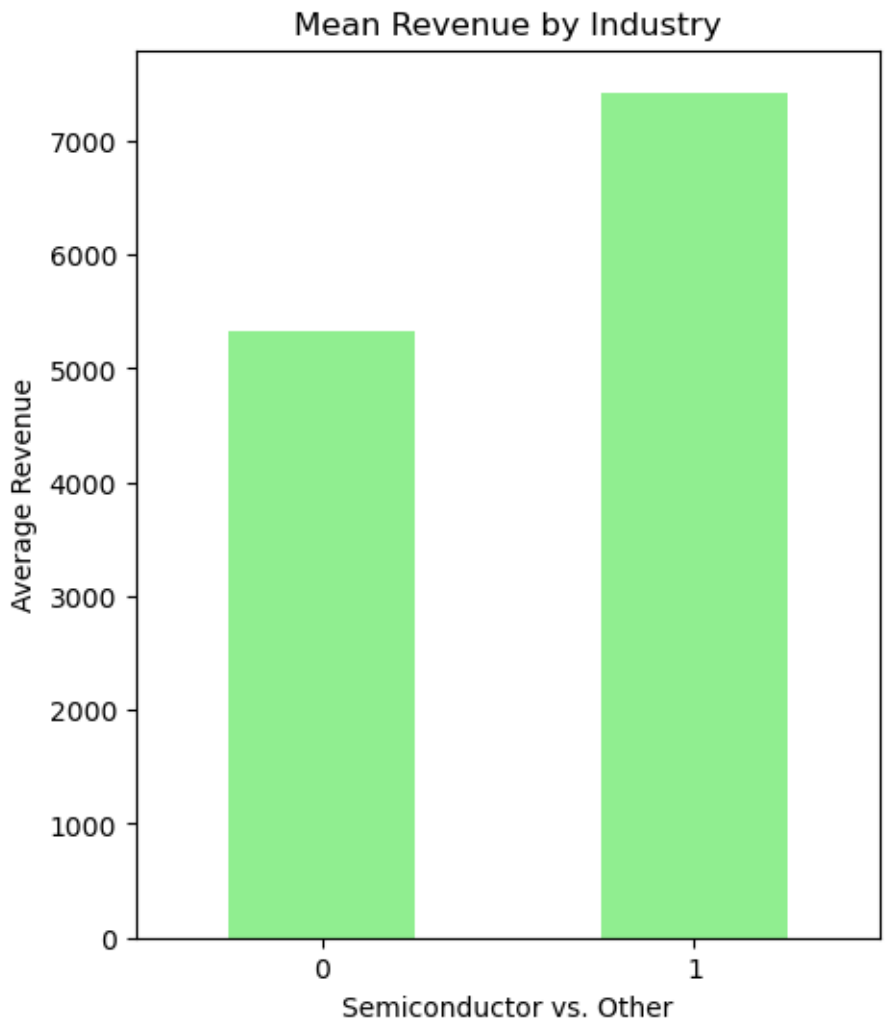
	car	log_revt	semiconductor	electronics	revt
count	5662.000000	5247.000000	5667.000000	5667.000000	5261.000000
mean	0.009801	4.937613	0.007235	0.016234	5341.599936
std	0.185124	2.942916	0.084757	0.126387	23414.697547
min	-1.375518	-1.234432	0.000000	0.000000	-11591.578000
25%	-0.066078	3.147197	0.000000	0.000000	44.532000
50%	-0.007698	5.321418	0.000000	0.000000	406.220000
75%	0.066937	7.019307	0.000000	0.000000	2225.659000
max	3.473246	12.560180	1.000000	1.000000	569962.000000

The descriptive statistics provide various statistics such as the three quartiles, minimum, maximum values, mean, standard deviation and the count. With the help of this data one can analyze easily about the skewness of their data and what are contributing factors that affect the interested industry.

4-2. The trend of the interested variable:

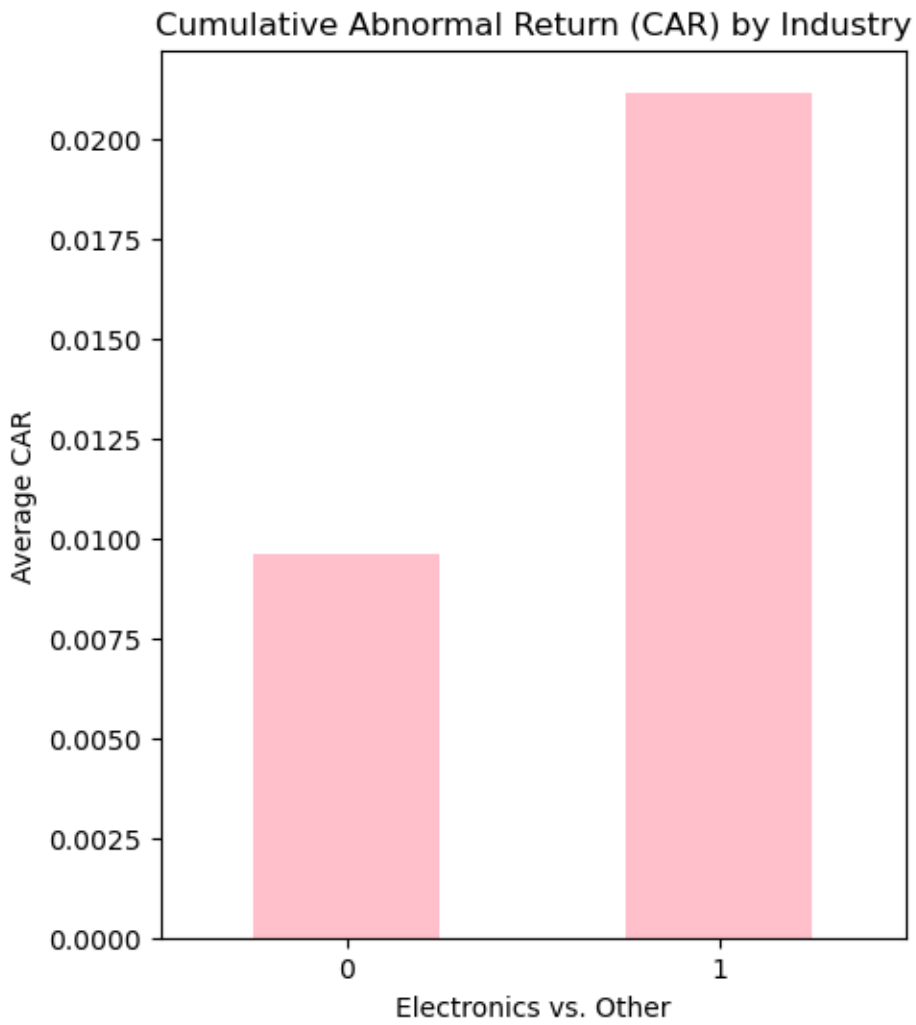
Semiconductor industry.

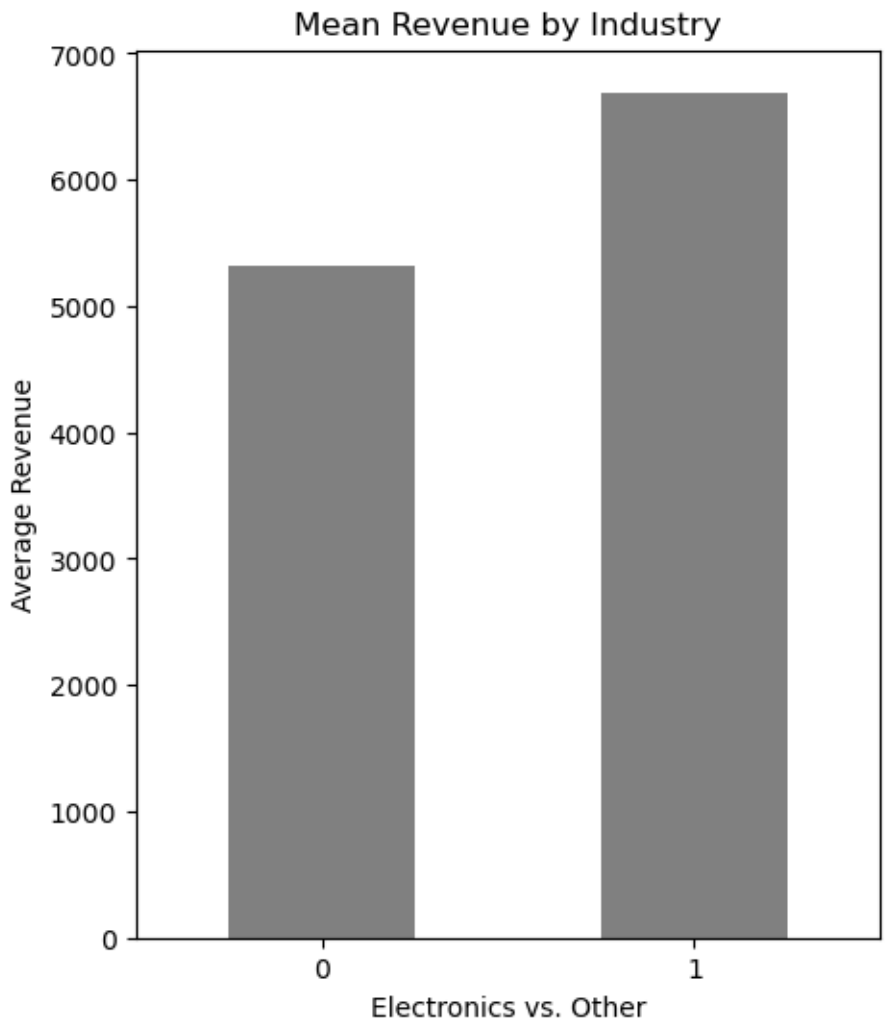

Cumulative Abnormal Return (CAR) by Industry
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
Average CAR
0
1
Semiconductor vs. Other



The semiconductor industry is performing well in both the car as well as the revenue, this indicates that the hypothesis is accurate, and the industry is performing well in the stock market as well as financial aspect of it as well. This shows that the industry has good potential going forward.

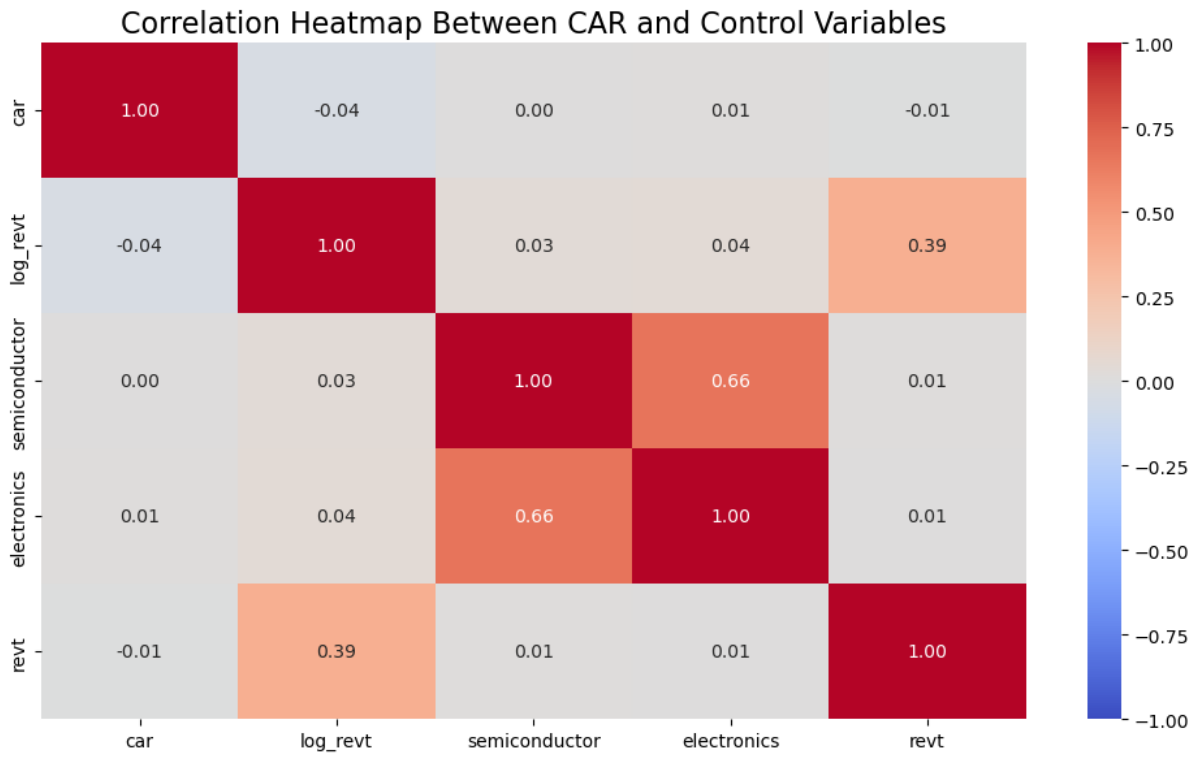
Electronics components not classified elsewhere:


Cumulative Abnormal Return (CAR) by Industry
Average CAR
0.0200
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0
1
Electronics vs. Other



Concerning the graphs provided, the industry electronics components not elsewhere classified are performing better in both revenue as well as the cumulative abnormal return as predicted by the hypothesis. Due to chips, there is a need of electronics components and due to their potential growth, the industry is performing well in the stock market as well.

4-3. Correlation



With reference to the above analysis, the chart shows a correlation between cars and other control variables such as the revenue, log revenue, and the interested industry variables such as semiconductor and electronics. Between the positive and negative there is a varying degree of correlation. And the highlighted part in red indicates the perfect positive relationship.

4.4 – Regression Analysis:

Q. Definition of regression analysis:

Regression analysis is nothing, but a statistical technique used to examine the connection between two or more variables in a way that allows for predicting or explaining one of the variables based on information about the others. It is the most used and utilized method to forecast and form conclusions.

Reference:

https://www.sciencedirect.com/topics/mathematics/regression-analysis

Definition of statistical terms:

Adj. R-squared: This is the modified version of R-squared which is adjusted for the number of variables in the regression. It increases only when an additional variable adds to the explanatory power of the regression.

Prob(F-Statistic): This tells the overall significance of the regression. This is to assess the significance level of all the variables together unlike the t-statistic that measures it for individual variables. The null hypothesis under this is “all the regression coefficients are equal to zero”. Prob(F-statistics) depicts the probability of the null hypothesis being true. As per the above results, the probability is close to zero. This implies that overall, the regressions are meaningful.

AIC/BIC: It stands for Akaike’s Information Criteria and is used for model selection. It penalizes the error mode in case a new variable is added to the regression equation. It is calculated as the number of parameters minus the likelihood of the overall model. A lower AIC implies a better model. Whereas BIC stands for Bayesian information criteria and is a variant of AIC where penalties are made more severe.

Prob (Omnibus): One of the assumptions of OLS is that the errors are normally distributed. An Omnibus test is performed to check this. Here, the null hypothesis is that the errors are normally distributed. Prob (Omnibus) is supposed to be close to the 1 for it to satisfy the OLS assumption. In this case, Prob (Omnibus) is 0.00, which implies that the OLS assumption is satisfied. Due to this, the coefficients estimated out of it are not Best Linear Unbiased Estimators (BLUE).

Durbin-Watson: Another assumption of OLS is homoscedasticity. This implies that the variance of errors is constant. A value between 1 to 2 is preferred. Here, it is 2.1 implying that the regression results are not reliable but still acceptable from the interpretation side of this metric.

Prob (Jarque-Bera): It is in line with the Omnibus test. It is also performed for the distribution analysis of the regression errors. It is supposed to agree with the results of the Omnibus test. A large value of the JB test indicates that the errors are not normally distributed.

Reference: CLASS NOTES - 9

http://localhost:8890/notebooks/Kubra/Downloads/in-class9_24F_shared.ipynb

Summary of findings for all the analysis: -

Model 1:

This analysis or interpretation is based on model one, the car is with the semiconductor industry and the r squared value being 0 suggests that the model has no variance whatsoever. The probability f statistic and f statistic value suggest that the model is not significant. The coefficient of the semiconductor suggests that there is a 0.0075 increase in the car.

Model 2:

This analysis or interpretation is based on model two, the car is with the electronics industry and the r-squared value being 0 suggests that the model has no variance whatsoever. The probability f statistic and f statistic value suggest that the model is not significant. The coefficient of the semiconductor suggests that there is a 0.0116 increase in the car.

Model 3:

This analysis or interpretation is based on model three, the car is with the semiconductor industry as well as the logarithmic revenue, and the r squared value being 0.2 percent suggests that the model has some variance. The probability f statistic and f statistic value suggest that the model is significant. As the p-value is less than 0.1 we can say that it is significant. The coefficient of the semiconductor suggests that there is a 0.0084 increase in the car.

Model 4:

This analysis or interpretation is based on model four, the car is with the electronics industry as well as the logarithmic revenue and the r squared value being 0.2 percent suggests that the model has some variance. The probability f statistic and f statistic value suggest that the model is significant. As the p-

value is less than 0.1, we can say that it is significant. The coefficient of the semiconductor suggests that there is a 0.0121 increase in the car.

Model 5:

This analysis or interpretation is based on model five, R squared value is 2.6% of the variation of the car. the f statistics and the p-value which is less than 0.1 suggest that the model is significant. The model includes various sic codes most of which do not show any significant effects on car.

Model 6:

This analysis or interpretation is based on model six, R squared value is 2.6% of the variation of the car. the f statistics and the p-value which is less than 0.1 suggest that the model is significant. The model includes various sic codes most of which do not show any significant effects on car.

Model 7:

This analysis or interpretation is based on model seven, the r squared value 0.1% if the variability of the car. The p-value is greater than 0.1 which suggests that the model as a whole is not significant. The coefficient of the semiconductor suggests that there is a 0.0083 increase in the car.

Model 8:

This analysis or interpretation is based on model eight, the r squared value 0.1% if the variability of the car. The p-value is greater than 0.1 which suggests that the model as a whole is not significant. The coefficient of the semiconductor suggests that there is a 0.0229 increase in the car.

5. Conclusion

5-1. Business implications:

As part of the Clark and co company, we have learned that due to the chips act event both the semiconductor and the electronics components not elsewhere classified industry clients are going to benefit as the event has a positive impact on these industries, choosing the model with the logarithmic revenue proves our hypothesis to be right as the p-value is less than 0.1 our model is significant and that there is a positive impact on the industry to the chips act event and our analysis can back this.

5-2: limitations of this research:

The data cannot be comprehensive such that the car might be influenced by various external factors such as market conditions, recession, and various other factors beyond the Chips Act. If the data does not consider all the factors, it is a simplified result. There also can be a bias that is if the industry was already performing well before the Chips Act, then it is difficult to establish the positive impact due to the event chips act.

5-3. potential project

Definition of predictive and prescriptive analytics: -

Predictive analytics helps companies make sense of the potential outcome. This can be achieved by using mined data, history, and statistics.

Prescriptive analytics looks at the future scenarios that is it takes a deeper look into what and why of a potential future outcome. It also helps a company make informed decisions.

References:

https://www.businessnewsdaily.com/8655-predictive-vs-prescriptive-analytics.html

Research question semiconductor:

Predictive: How will the CHIPS and Science Act affect the semiconductor industry?

Prescriptive: What strategies should the semiconductor companies adopt to maximize the benefits of the CHIPS and Science Act?

Research question electronics:

Predictive: How will improvements in production efficiency due to the CHIPS Act impact the stock market performance of the companies in the electronics components industry?

Prescriptive: What course of action should the companies in the electronics components industry implement to optimize the benefits of the CHIPS Act?